SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I N. Chawran Aryan (192365053) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

N.Chawran Aryan
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

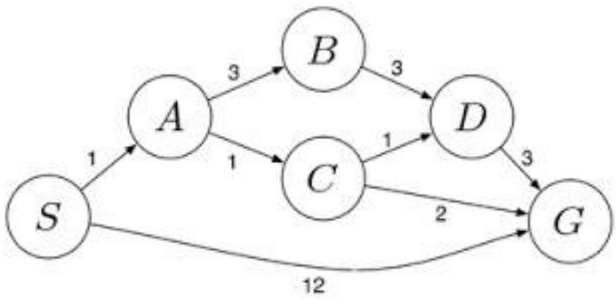
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. Water Jug Problem (4-Gallon and 3-Gallon)
Initial State

- (0,0) → Both jugs are empty.

Goal State

- (2,0) → Exactly **2 gallons** in the 4-gallon jug.

Solution Steps

Step	4-Gallon Jug	3-Gallon Jug	Action
1	0	0	Initial state
2	0	3	Fill 3-gallon jug
3	3	0	Pour into 4-gallon jug
4	3	3	Fill 3-gallon jug again
5	4	2	Pour into 4-gallon until full
6	0	2	Empty 4-gallon jug
7	2	0	Pour remaining 2 gallons into 4-gallon jug

Final Answer
The **4-gallon jug contains exactly 2 gallons**, which satisfies the goal.

2. Mars Rover Intelligent Agent
(i) Percepts

- Camera images
- Rock and soil samples
- Temperature
- Atmospheric pressure
- Chemical composition
- Distance from obstacles
- Wheel position and movement

(ii) Operating Environment

- **Partially Observable**
- **Dynamic**
- **Sequential**
- **Continuous**
- **Single Agent**
- **Unknown environment**

(iii) Actions

- Move forward/backward
- Turn left/right
- Capture images
- Collect rock and soil samples
- Analyze samples
- Send data to Earth
- Avoid obstacles

(iv) Performance Measure

- Accuracy of collected data
- Number of successful samples collected
- Safe navigation
- Battery efficiency
- Mission completion

(v) Suitable Agent Architecture
Model-Based Goal-Based Agent
Reason:

- Maintains an internal model of the environment.
- Plans routes toward scientific goals.
- Makes decisions even when all information is not available.

3. 8-Queens Problem Formulation
Initial State
An empty 8 × 8 chessboard.
State Space
All possible arrangements of eight queens on the board.
Operators
Place one queen in a safe position in each column.
Goal Test
No two queens attack each other:
- Same row
- Same column
- Same diagonal

Path Cost
Each queen placement has equal cost.
Search Strategy
- Backtracking Search
- Depth First Search (DFS)

Example Solution
```
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```
This arrangement satisfies all constraints.

4. OLA Cab Booking Problem
Type of Problem-Solving Agent
Goal-Based Agent
Reason
The customer wants to reach a destination by selecting the most suitable cab based on comfort and availability.
Pseudocode
```
START

Input pickup location
Input destination

Display available cab types
(Mini, Micro, Sedan, Prime, Shared)

User selects cab

IF selected cab is available
    Book cab
    Calculate fare
    Display driver details
ELSE
    Display "Cab Not Available"
ENDIF

END
```

5. Uniform Cost Search (UCS)

Algorithm

UniformCostSearch(Graph, Start, Goal)

1. Insert Start node into priority queue with cost = 0.
2. Mark Start as visited.
3. While queue is not empty:
   Remove node with minimum cost.
   If node is Goal:
      Return solution.
   Expand current node.
   Add neighboring nodes with cumulative cost.
   Update queue if lower-cost path is found.
4. If Goal is not found:
   Return Failure.

Working

- Starts from source node **S**.
- Expands the node with the **lowest cumulative cost** first.
- Continues until destination **G** is reached.
- Guarantees the **least-cost path**.

Advantages

- Complete algorithm.
- Always finds the optimal path when costs are non-negative.

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